C - 10AM

11:30

~~APTI~~

Backup-C → VPL - on LMS

HackerRank

Today - NO 2nd sessin

Tomorrow - C-Backup - VPL.

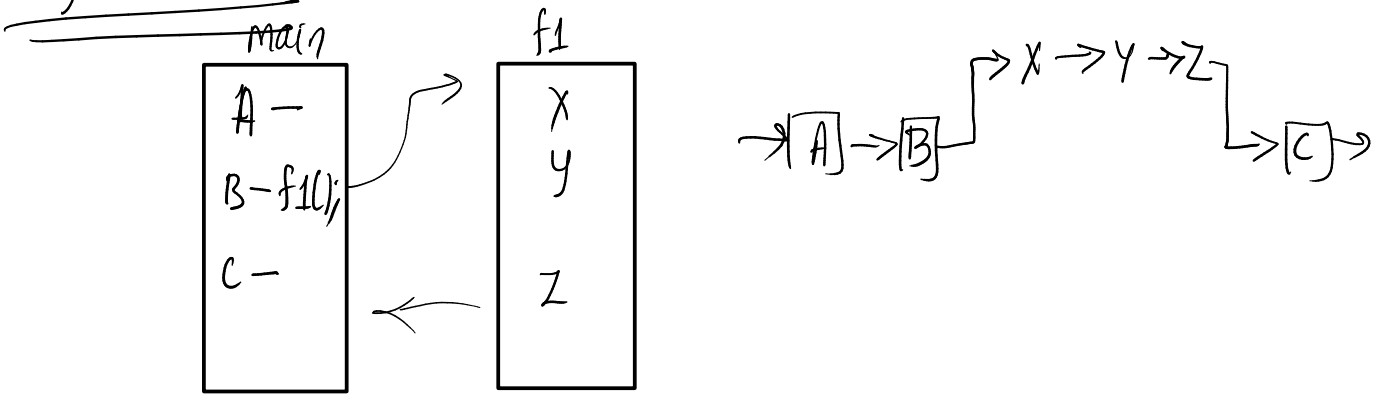
Mon - wed - Fri - Apti

Tue thu sat - C Backup

Control statements

→ stmt which affects, the flow control in a program.

Regular flow



control stmt → Iterative / loop / Repeatative : while, do-while, for

control stmt → Non-Iterative : → with condition : if, if-else, switch-case

→ Without condition : goto, return, break, continue

Syntax of Break:

```
break;
```

→ only allowed inside loop body.

→ or inside switch.

```
main() {
    Printf("Hello\n");
    break; //Error.
    Printf("Bye\n");
}
```

continue:

```
continue;
```

→ MUST be inside loop body.

return:

```
return;
return data;
```

returns the control (sometimes with data) to the calling function

```
main()
{
A -
B -
C f1();
D -
E -
}
calling fn
```

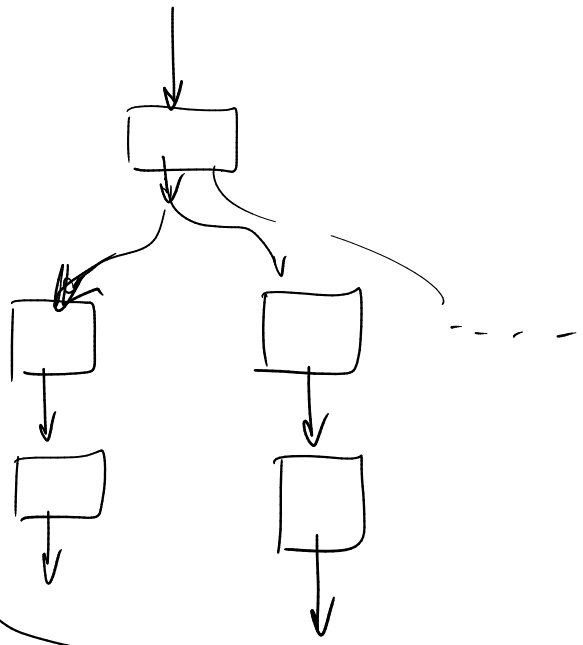
```
f1()
{
  X -
  return;
  Z;
}
called fn
```

return: from function → to calling function | break the loop | continue the loop

goto a LABEL ⇒ Jump within the function block

```
main() {
    A: Pf("first \n");
       Pf("2nd \n");
       goto B;
       printf("3rd \n");
    B: printf("4th \n");
       goto C;
       Pf("5th \n");
    C: Return 0;
}
```

Branching



if...else or Switch...case

To check status of a (bit) in a (data):

data = data & (1 << bit);

data: 32 bit

bit: 5

status = data & (1<<bit);

\Bitmask

data = 50

bit = 4

Pf("%d\n", data & (1<<bit)); // 16

(data>>bit)&1 // 1

data & (1<<bit) => 0 or Nonzero (2^{bit})

(data>>bit) & 1 => 0 / 1

x

bit = 2;

"%d", data & (1<<bit))

(data>>bit)&1

data = 50 => 0 0 1 1 0 0 1 0 (7 6 5 4 3 2 1 0)

data>>bit

&

1 0 0 0 0 0 0 0 1

result = 0 0 0 0 0 0 0 0

bit

1<<2 : 0 0 0 0 0 1 0 0 (7 6 5 4 3 2 1 0)

data = 50 : 0 0 1 1 0 0 1 0

0 0 0 0 0 0 0 0

result = data & (1<<bit);

data = 50 : 0 0 1 1 0 0 1 0

data>>2

&

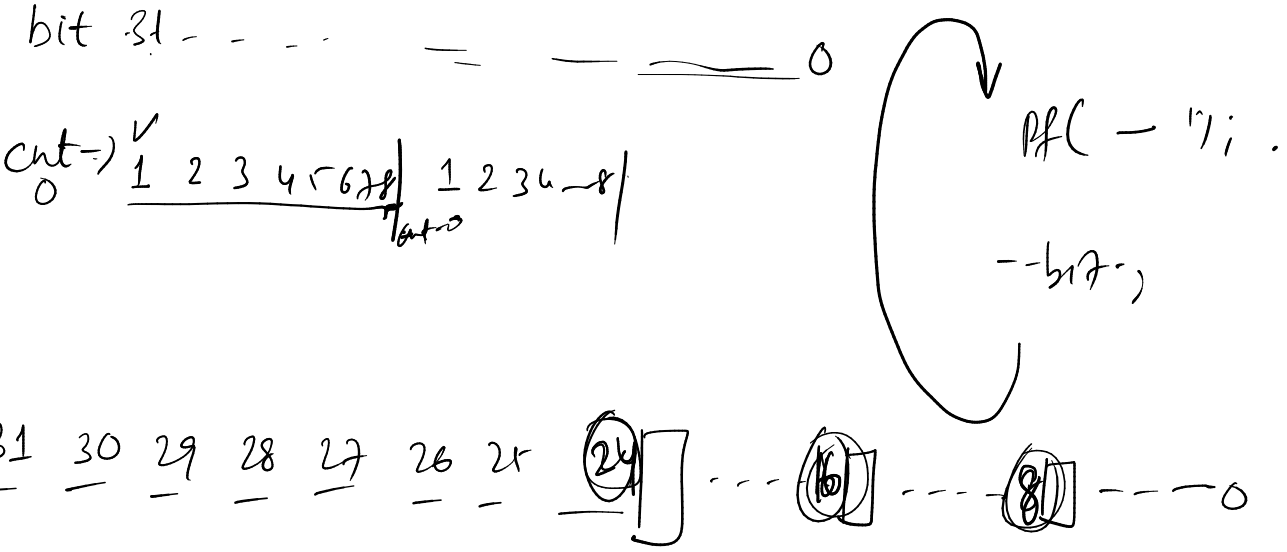
1 0 0 0 0 0 0 0 1

result = 0 0 0 0 0 0 0 0

1 & x => 1 / 0

bit => 31 30 29 28 27 ..... 2 1 0 | -1

cnt=0 bit = 31



bit =) 31 30 29 28 27 26 25 24 ... 16 ... 8 ... 0

if ((bit == 24) || (bit == 16) || (bit == 8))

Pf(1);

Assignment-1:

W.A.P to count the No. of bits set, in an integer.

I/P	O/P
0	0
-1	32
1024	1
50	3

2nd: count the set-Bit-Pairs in an integer

I/P	O/P
50	1
00110010	
127	3
01111111	
-1	16

3rd : Find the longest series of set Bits in an Integer